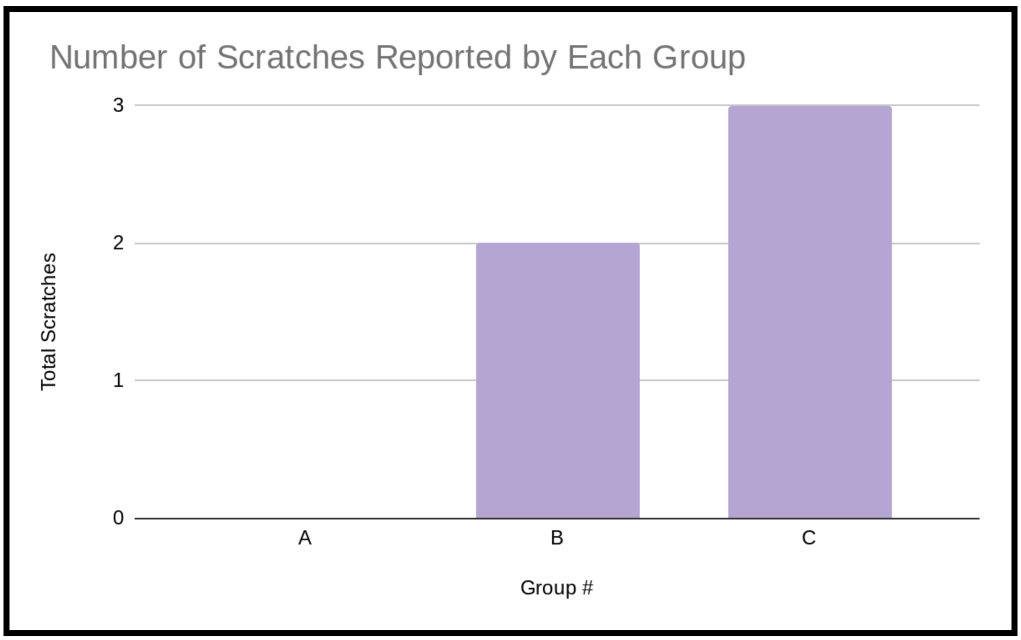
Element I: Test Results and Analysis

After following the plan laid out in the Scratch Test, we have documented these results.

Group A	Scratches reported	Group B	Scratches Reported	Group C	Scratches reported
#1	0	#6	1	#11	2
#2	0	#7	0	#12	0
#3	0	#8	0	#13	1
#4	0	#9	1	#14	0
#5	0	#10	0	#15	0



-There was a total of 15 adult participants, and 4 of them had reported scratches. An outlier in our data is that one person had reported a total of 2 scratches. While inspecting our projects made with Makey Makey more carefully, we have discovered where the scratch points were and have documented that, so we know to file that edge in the future. Overall, we are proud that we did not run into more scratches during our test.

After performing the “Throw” Test, we have documented the following results.

Group By Strength	Did it Break?
Weak #1	No
Weak #2	No
Average #1	No
Average #2	Yes
Strong #1	No
Strong #2	Yes

-To carry out this test, we have gathered 6 individuals, all with varying strength levels. We had also bought 6 more copies of the Makey Makey kit so that no kit was being thrown more than once. To be allowed in the weak category, you would have to be unable to lift more than 50 pounds. For this, we had gathered two female classmates, and they each slammed the kits against our concrete floor. Next, for our average throwers, you cannot be able to lift more than 100 pounds and be able to lift at least 50 pounds to be considered for this group. For this, we have gathered one male and one female adult teacher. While the male teacher’s kit did break, the female teacher’s did not. While the kit did not completely break in half, it did, however, fracture a piece of the kit. Finally we gathered our two strong individuals. To be considered for this class, you had to be unable to lift more than 200 pounds but also able to lift more than 100 pounds. In this study, our two strong individuals were two males who work in construction. While the first man’s kit did not break or fracture, the second one did. The second man, upon throwing his kit to the ground, had documented that it completely broke in half. While it is still impressive that we only had one complete break and one fracture, we still want to experience zero breaks in the future. Because of this, we have written that the kits need to build stronger inside walls.

After performing the Stress Test, we have documented these results.

Weight (In Pounds)	Time	Breakage?
5	10 Minutes	No
10	20 Minutes	No
15	30 Minutes	No

To perform the stress tests, we have bought more kits to test out. To perform this test, we gathered 3 circular weights that would lie across each kit. We had decided that a fair way to test the effectiveness of the durability of the kits is to not only increase the weight each time, but also the time the weight stays on. As documented above, the 5-pound weight stayed on the kit for 10 minutes, the 10-pound weight stayed on for 20 minutes, and the 15-pound weight stayed on for 30 minutes. While originally we did not plan to use a multiple of two for weights to minutes, it just ended up working out that way. Needless to say, we were shocked that the kit did not collapse under the 15-pound weight, especially since it was on there for 30 minutes. I will say that overall, our weight durability is good; however, due to the previous test, we are still planning to improve upon it.

After performing the Instructions Test, we have documented these results

Control Group X	Control Group # of Kids	Instructions	Did they follow the instructions?
A	10	Only Written	0/10 did
B	10	Written and Demonstrated	6/10 did
C	10	Written and song	8/10 did

-For this final test, we decided to test what method would work best for actually teaching the kids how to use the Makey Makey kits. To keep this test as fair as we possibly could, we gathered 3 groups of ten kids, where the average age in the group was 5. Each group of ten also consisted of an even number of boys to girls, so 5 boys and 5 girls in each group. We also made sure that all 30 kids had no prior knowledge of using these kits. Control Group A was given only a set of written instructions on how to use the kits while we observed. A lot of difficulty was seen among this group of kids, and none of them left the room able to follow the instructions. Next, Control Group B was sent in with a group of written instructions and a live demonstration, and we observed the following. While this group struggled in the beginning, 6 of them were able to leave with being able to use the kit. While the success rate for Group B was 60%, we still wanted to see if we could get that number improved in the next control group. Finally, for Control Group C, we decided to go with a ‘How To Use Makey Makey Song’ as well as set of written instructions. After letting them learn the song, we decided to give them the prototypes and observe the results. With the song, the kids took to tying the prototype rather quickly as they would sing the song each time they went to use the kit. While Group C’s success rate was not 100%, it was vastly improved upon Group B’s results. Because of this, we have determined that Group C’s method was the best for teachers to implement into their classrooms.

In short, I would say our tests met the results that we were looking for in our design. As for the durability of our product, I would say we hit the mark. While it is not 100% perfect, as documented above, I would say our most successful test was that of determining which method was the best to teach each kid, and was overall surprised that the best method was the song and written instructions. I would say overall that our tests were a huge success.